

# Synthesis and in vivo evaluation of novel benzimidazole-sulfonamide hybrids and *Lucilia cuprina* maggots' excretion/secretion topical gels for wound healing

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## Abstract

A novel series of benzimidazole-sulfonamide hybrids (**6a–g**) was designed, synthesized, and their structures were characterized by <sup>1</sup>H NMR and <sup>13</sup>C NMR spectral data. Further, all compounds were evaluated for their antibacterial effects. The synthesized benzimidazole-sulfonamide hybrid (**6d**) showed the highest antibacterial activity against *Staphylococcus aureus*, *Staphylococcus epidermis*, *Escherichia coli*, and *Klebsiella pneumonia*. Different gel formulations for **6d** were prepared and physically investigated. After in vivo evaluation for wound healing, the **6d** gel formulation delayed the wound healing in animal model. But interestingly, a combination of **6d** and *Lucilia cuprina* maggots' excretion/secretion showed remarkable enhancement in wound healing compared with **6d** only. This combination opens up promising prospects for wound healing formulations.

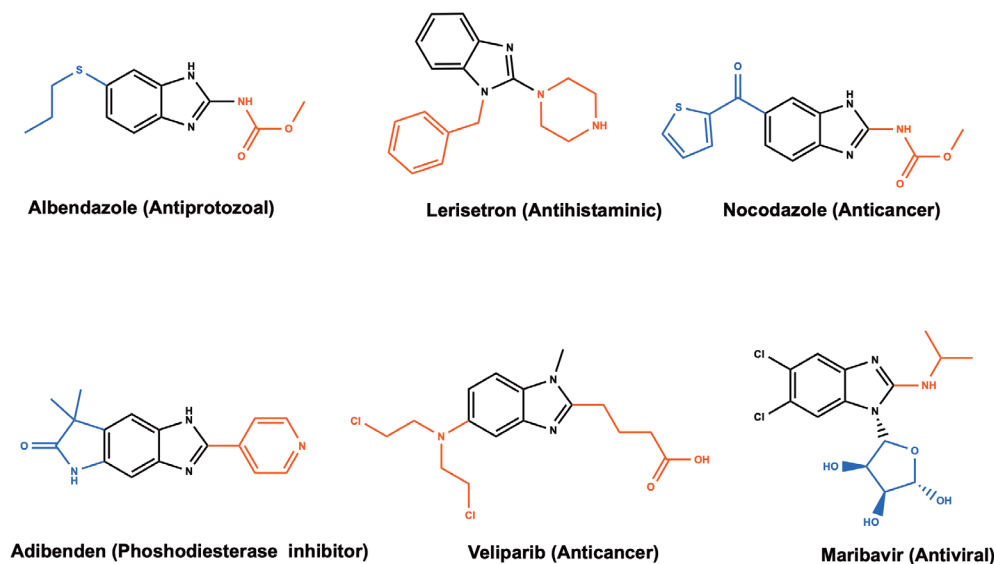
## KEYWORDS

antibacterial, benzimidazole, excretion/secretion, gel formulations, *Lucilia cuprina*, sulfonamide, wound healing

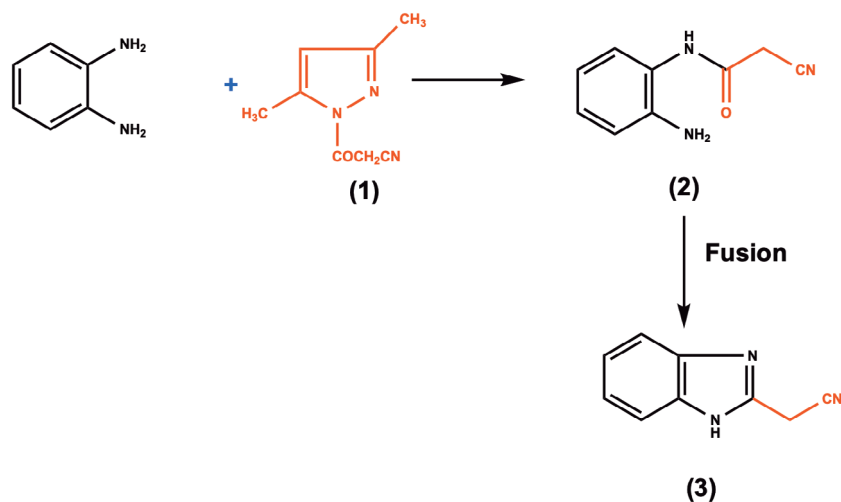
## 1 | INTRODUCTION

Synthesis and evaluation of hybrid molecules as a variety for pharmacological agents and effective drugs has gradually escalated.<sup>[1,2]</sup> Benzimidazole derivatives are a building block of many bioactive compounds, which have been considered an important class of pharmacologically active analogs in pharmaceutical and medicinal chemistry.<sup>[3–6]</sup> Heterocycles having benzimidazole rings are also of considerable interest because of their synthetic utility as synthetic reagents in multicomponent reactions.

In addition, there are well-known natural products containing benzimidazole moieties that have various physiological, pharmacological, and toxic properties.<sup>[7,8]</sup> In recent years, several drugs have been developed from imidazole derivatives. For example, Albendazole (Antiprotozoal); Lerisetron (Antihistaminic), Nocodazole (Anticancer); Adibenden (Phosphodiesterase inhibitor); Veliparib (Anticancer); Maribavir (Antiviral), Figure 1.<sup>[9]</sup> Sulfonamides are widely used as antimicrobial,<sup>[10,11]</sup> anti-inflammatory,<sup>[12]</sup> anti-obesity,<sup>[13]</sup> diuretic,<sup>[14]</sup> hypoglycemic,<sup>[15]</sup> antithyroid,<sup>[16]</sup> anticancer,<sup>[17,18]</sup> and antiviral



**FIGURE 1** Some marketed drugs containing benzimidazole nucleus



agents as well as anti-neuropathic pain.<sup>[19]</sup> Some sulfonamide derivatives are also well recognized as ulcerative colitis,<sup>[20]</sup> rheumatoid arthritis,<sup>[21]</sup> and male erectile dysfunction as the phosphodiesterase-5 inhibitor sildenafil.<sup>[22]</sup> Pathogenic organisms such as bacteria can easily penetrate the body through the wounds, which become contaminated and delayed in the healing. So, developing the wound healing agents especially from natural sources is a critical issue. As a part of our on-going research in systematic investigation of synthesizing some novel bioactive compounds based on literatures,<sup>[23–26]</sup> here we designed benzimidazole linked sulfonamide hybrid molecules and extracted *Lucilia cuprina* maggots' excretion/secretion with the aim of developing a new class of antimicrobial agents to be used in topical wound healing.

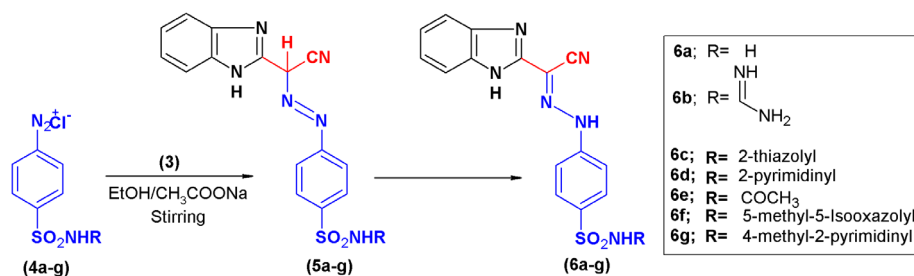
## 2 | RESULTS AND DISCUSSION

### 2.1 | Chemistry

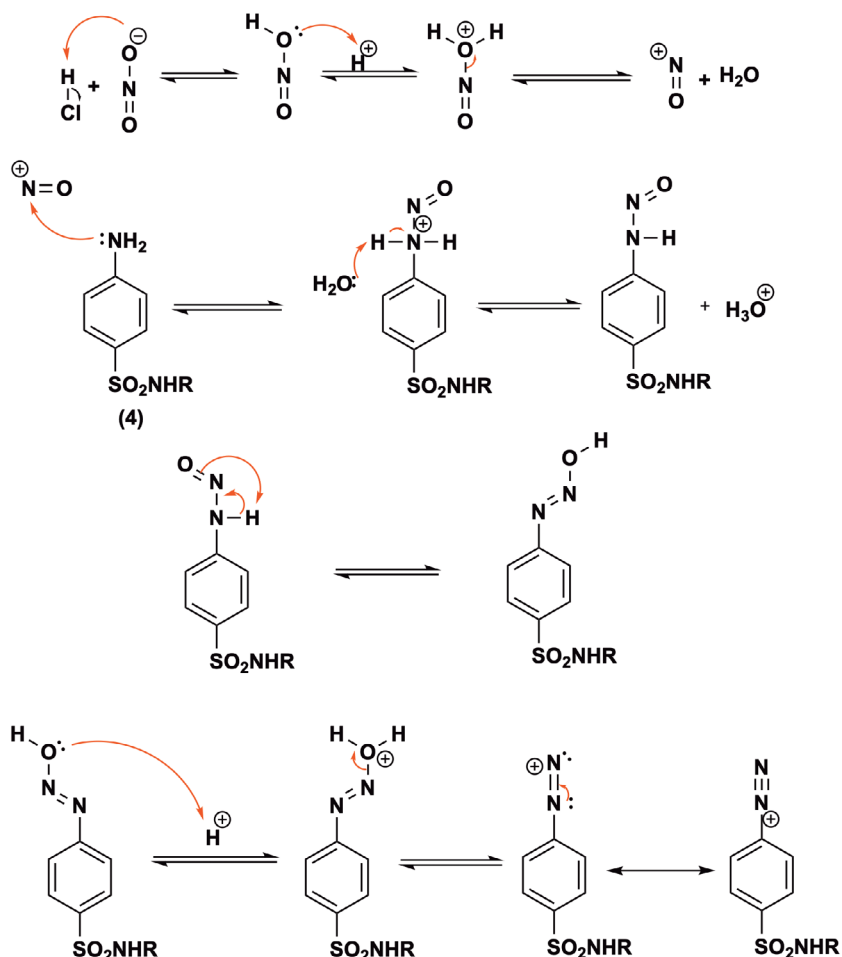
The starting 2-cyanomethylbenzimidazole (**3**) was obtained in high yield from the trans mono cyanoacetylation of o-phenylenediamine using 1-cyanoacetyl-3,5-dimethylpyrazole (**1**) as the acetylating agent followed by thermal fusion of *N*-(2-aminophenyl)-2-cyanoacetamide (**2**) according to a published procedure<sup>[27]</sup> (Scheme 1).

A series of *N*-(4-sulfamoylphenyl)-1*H*-benzo[*d*]imidazole-2-carbo hydrazonoyl cyanide derivatives (**6a–g**) were obtained via treatment of 2-cyanomethylbenzimidazole (**3**) with diazonium salt of sulfonamide derivatives (**1a–f**) at room temperature in ethanol in presence of sodium acetate, according to the steps depicted in (Schemes 2–4).

**SCHEME 2** Coupling reaction of diazonium electrophile with 2-cyanomethyl benzimidazole



**SCHEME 3** Generation of diazonium salts

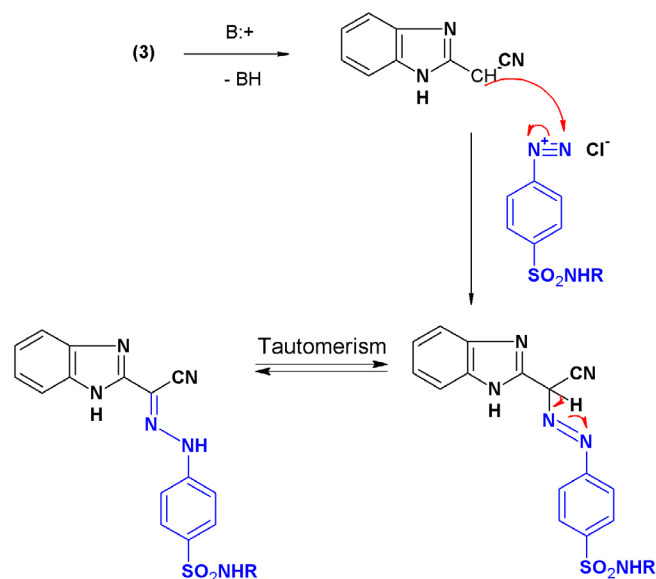


Diazotization was carried out in ethanol in the presence of sodium acetate and nitrosyl chloride at 0–5°C, (Scheme 2). Based on their analytical and spectral results, the structures of the synthesized compounds (**6a–g**) have been identified rather than the tautomeric form (**5**). The infrared spectra of compounds (**6a–g**) showed absorption bands characteristic for N=N at 1,518–1,624 cm<sup>−1</sup> range and C≡N at 2,214–2,233 cm<sup>−1</sup> range functional groups. The representative <sup>1</sup>H NMR spectrum of compound (**6a**) (DMSO-*d*<sub>6</sub>) revealed the lack of singlet characteristic for methine proton and exhibited a sharp singlet signal at δ 5.40 ppm referring to the amino proton, a broad singlet at δ 7.35 ppm due to the NH protons and showed downfield

signal at δ 14.71 ppm assigned to the NH proton of benzimidazole moiety in addition to the presence of aromatic protons at δ 7.61–7.92 ppm. In addition, in the <sup>1</sup>H NMR spectrum of compound **6c** we observed two singlet signals at δ 5.29, 5.31 ppm assigned to the thiazole protons and showed downfield signal at δ 14.68 ppm assigned to the NH proton of benzimidazole moiety and a multiplet at δ 7.26–7.68 ppm assigned to the aromatic and NH protons. The coupling reactions involved that initially generate strong N<sup>2+</sup> electrophiles from different amines then finally react and coupled at methylene group to produce *N*-(4-substituted sulfamoyl phenyl)-1*H*-benzo[*d*]imidazole-2-carbohydrazonoyl cyanide derivatives (**6a–g**) (Scheme 2).

## 2.2 | Solvatochromic study of benzimidazole-sulfonamide hybrids

The solvent effects of the novel benzimidazoles-sulfonamide hybrid were studied on UV-Visible spectrophotometer at room temperature. The absorption spectra of the (6a-g) compounds were reported at  $10^{-5}$  M in dimethyl formamide (DMF) and chloroform (CF) and represented in Figure 2. All synthesized compounds showed bathochromic shift in DMF as compared to CF. These bathochromic shifts can be attributed to the interaction of H atom of amino proton of synthesized compounds with polar aprotic solvent

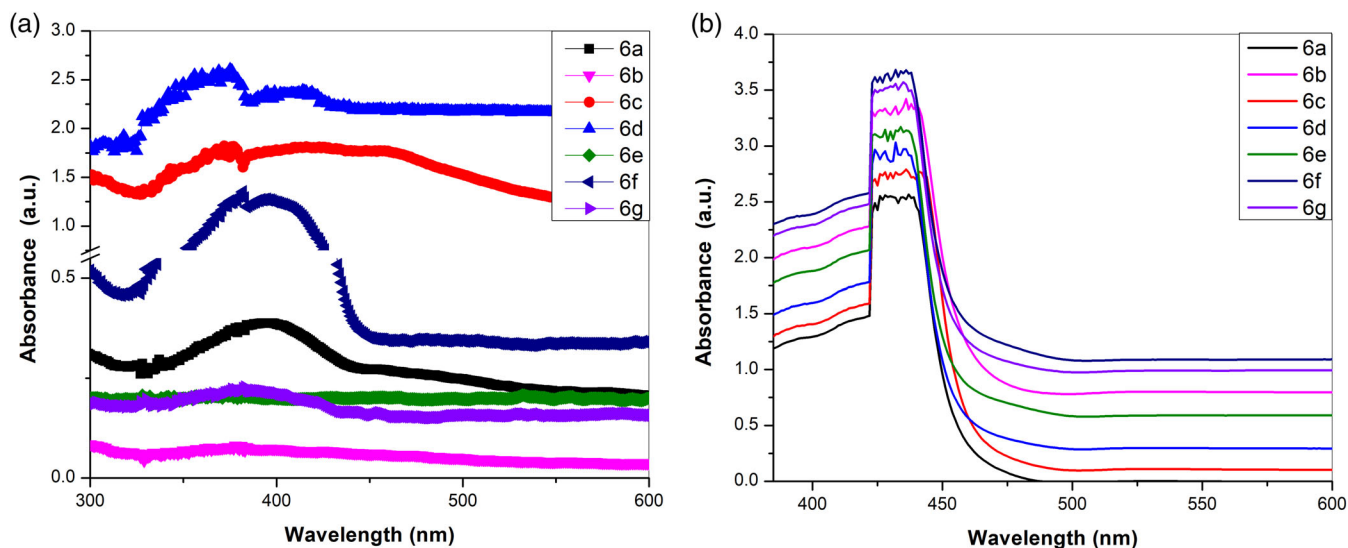


**SCHEME 4** Formation mechanism of benzimidazole-sulfonamide hybrids

because of increase polarity of the synthesized compounds system, generally in the excited state.

## 2.3 | Antibacterial activity

The novel synthesized series of *N*-(4-sulfamoylphenyl)-1*H*-benzo[*d*]imidazole-2-carbo hydrazonoyl cyanide derivatives (6a-g) were examined for their antibacterial activity against *Staphylococcus aureus* and *Staphylococcus epidermidis* (as Gram positive bacteria) and *Escherichia coli* and *Klebsiella pneumonia* (as Gram negative bacteria). The tested derivatives were showed varying antibacterial activities as showing in Table 1. Compound (6d) showed the highest effect against all tested bacteria among the other compounds. This might be due to the pyrimidinyl group that is a highly efficient pharmacophore. We can conclude that introduction of pyrimidinyl group into *N*-(4-substituted sulfamoyl phenyl)-1*H*-benzo[*d*]imidazole-2-carbo hydrazonoyl cyanide may enhance the antibacterial activity. This possibility was confirmed by displaying compound (6g) high antibacterial activity against *S. epidermidis*, which has 4-methyl 2-pyrimidinyl group. In the same time, compound (6b) showed high activity against *E. coli*, which has a NH=CH-NH<sub>2</sub> group. The other compounds did not show any activities against tested bacteria. In addition, compound (6d) was studied for minimum inhibitory concentration (MIC), where it showed 312.5, 156.25, 625, and 1,250 µg/ml for *S. aureus*, *S. epidermidis*, *E. coli*, and *K. pneumonia*, respectively. In previous work,<sup>[28]</sup> the MIC of compound 6d against *S. aureus* (ATCC-6538-P) was 75 µg/ml, which is lower than our results in this



**FIGURE 2** UV-Vis spectra of novel synthesized series of *N*-(4-sulfamoylphenyl)-1*H*-benzo[*d*]imidazole-2-carbo hydrazonoyl cyanide derivatives (6a-g) in CF (a) and DMF (b)

**TABLE 1** Antibacterial activities of novel synthesized sulfonamide derivatives

| Sample code | Gram positive bacteria        |                                    | Gram negative bacteria       |                                   |
|-------------|-------------------------------|------------------------------------|------------------------------|-----------------------------------|
|             | <i>S. aureus</i> (RCMB 10010) | <i>S. epidermidis</i> (RCMB 009-2) | <i>E. coli</i> (RCMB 010052) | <i>K. pneumoniae</i> (RCMB 003-1) |
| 6a          | —                             | —                                  | —                            | —                                 |
| 6b          | —                             | —                                  | 9.07 ± 0.21                  | —                                 |
| 6c          | —                             | —                                  | —                            | —                                 |
| 6d          | 11.97 ± 0.15                  | 13.03 ± 0.06                       | 13.93 ± 0.4                  | 10.1 ± 0.36                       |
| 6e          | —                             | —                                  | —                            | —                                 |
| 6f          | —                             | —                                  | —                            | —                                 |
| 6g          | —                             | 13.63 ± 0.35                       | —                            | —                                 |

**TABLE 2** Physical properties of gel formulations

| Gel formulation code | pH          | Spreadability (g cm s <sup>-1</sup> ) |                                | Viscosity (Poise) | Surface tension (dyne/cm <sup>2</sup> ) | Skin irritation |
|----------------------|-------------|---------------------------------------|--------------------------------|-------------------|-----------------------------------------|-----------------|
|                      |             | Weight (g)                            | S (g cm s <sup>-1</sup> )      |                   |                                         |                 |
| Blank 1              | 8.31<br>8.1 | 20                                    | 4.4                            | 662               | 691.1                                   | 0               |
| Blank 2              | 5.2         | —                                     | —                              | 1,880             | 463.3                                   |                 |
| Blank 3              | 6.6         | 20<br>30<br>50<br>100                 | 0.975<br>3.84<br>14.9<br>154.8 | 1,733             | 400.3                                   |                 |
| Blank 4              | 7.2         | 50                                    | 16.14                          | 1,678             | 462.4                                   |                 |
| Blank 5              | 6.4         | —                                     | —                              | 1,360             | 570.4                                   |                 |
| Blank 6              | 5.94        | 50                                    | 15.3                           | 1,970             | 635.8                                   |                 |
| Blank 7              | 5.3         | 50                                    | 2.59                           | 2,323             | 524.5                                   |                 |
| 6d gel               | 7.2         | 50                                    | 17.12                          | 1,085             | 368.9                                   |                 |
| ES gel               | 7.3         | 50                                    | 16.39                          | 1,111             | 386.5                                   |                 |
| 6d-ES gel            | 7.2         | 50                                    | 17.88                          | 1,053             | 402.4                                   |                 |

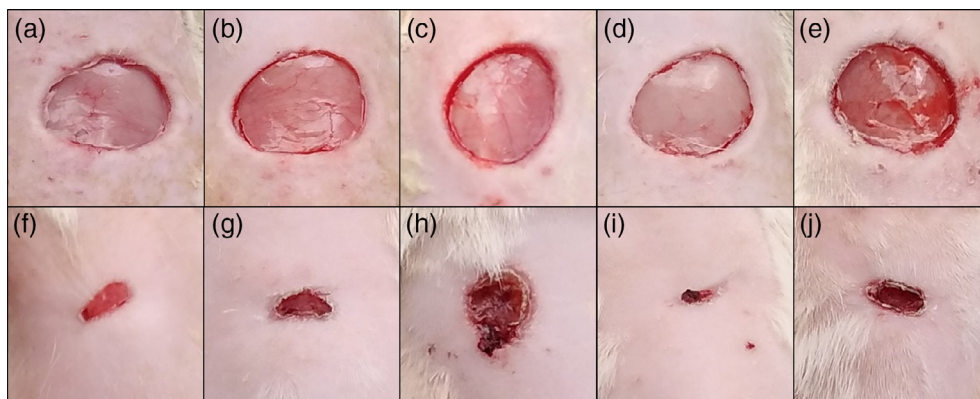
work. The difference comes from the difference of the strains. From the obtained results, the sulfonamide derivative (6d) showed highest activity with Gram +ve bacteria more than Gram -ve bacteria.<sup>[24-27,29]</sup>

## 2.4 | Physical characterization of gel formulation

A set of blank gels were successfully prepared using different concentrations of carbopol (CRB) (0.5, 1, 1.5, and 2% wt/wt) as gelling agent. To adjust a neutral pH in order to avoid a skin irritation, different concentrations of triethanolamine (TEA) (1, 1.5, and 2% wt/wt) were

examined. Then, the blank gels were comparatively evaluated for various characterization parameters, including pH, viscosity, skin irritation, surface tension, and spreadability. According to Table 2, pH values of the blank and gel formulations were in the close range of normal pH (5–7), which is considered acceptable to avoid skin irritation upon application. Spreadability is the expansion area of the gel formulation to spread for a specific application. Spreadability for the blank gels was evaluated using different weights tied to upper slide (20, 30, 50, and 100 g), the calculated results are represented in Table 2. The most suitable weight was 50 g with spreadability equals 14.9 g cm s<sup>-1</sup>, the values of spreadability indicate that the gel is easily spreadable by small amount of shear. The





**FIGURE 3** Wound healing activity of novel sulfonamide derivative gel. (a–e) the initial wound of control (a), –ve control (b), **6d** (c), ES (d), and **6d-ES** (e) groups. The wound contraction after 11 days for control (f), and after treatment with blank gel (g), **6d** gel (h), **6d-ES** gel (i), and ES gel (j)

viscosity of all gels was in the range of 662.5 to 2,323 P. As can be seen in Table 2, the concentration of polymer increases, the viscosity of the formulation consequently increases. In addition, the prepared gels were found to be homogeneous (no clog appearance) and did not develop any irritation upon application to the skin for 6 hr. Actually, the best sample was no. #4 with 1.5 ml TEA,  $\text{pH} \approx 7$  at 1% CRB in the presence of solvent. In addition, the increase the CRB concentration, the increase the gel's viscosity. While the increase the TEA concentration, the decrease the gel's viscosity. All gel formulations have identical physical properties as shown in Table 2.

## 2.5 | Wound healing activity

As seen in Figure 3, the wound healing activity of novel sulfonamide derivative (**6d**) in a gel formulation was observed in rats (excision wound, 25 mm in diameter) over time. The wound contraction % increased gradually. After 11 days, the contraction % achieved  $90.88 \pm 1.16$ ,  $89.99 \pm 1.50$ , and  $73.71 \pm 3.38\%$  for control, –ve control, and treated groups, respectively. The delay of the wound healing process with sulfonamide derivative treatment may be due to the exudation of tissue fluids,<sup>[30]</sup> which leads to increase the time of wound healing process.<sup>[31,32]</sup> Sulfonamide derivatives can be used for reducing the wound infection.<sup>[33]</sup>

By combination with other compounds that have wound healing activities such as chitosan and excretion/secretion (ES) of medicinal maggots, the sulfonamide derivatives may show synergistic antibacterial and wound healing activities.<sup>[29]</sup> Sulfonamide derivative (**6d**) was investigated with ES of *L. cuprina* maggots for wound healing activity. It showed an increase in the wound contraction of  $83.01 \pm 1.12\%$ , while the ES only showed  $95.33 \pm 2.24\%$ . ES of *L. cuprina* maggots increases the

wound healing<sup>[34–36]</sup> by boost the fibroblast cells migration and tissues formation.<sup>[37–40]</sup>

## 3 | EXPERIMENTAL

### 3.1 | Chemicals and instruments

All chemicals used in this study were of analytical reagent grade and of highest purity available. Organic solvents were purchased from British Drug House. Melting points ( $^{\circ}\text{C}$ , uncorrected) were determined in the open capillaries on a Gallenkamp melting point apparatus (Sanyo Gallenkamp, Southborough, United Kingdom). IR spectra (KBr disks) were measured on an FTIR plus 460 or Pye Unicam SP-1000 spectrophotometer (Pye Unicam, Cambridge, United Kingdom). NMR spectra ( $\text{DMSO}-d_6$ ) were recorded on a Bruker spectrophotometer (400 MHz for  $^1\text{H}$  NMR and 100 MHz for  $^{13}\text{C}$  NMR). Using  $\text{DMSO}-d_6$  as a solvent and tetraethylsilane as internal standard chemical shifts are expressed as  $\delta$  ppm. Perkin Elmer lambda 35 spectrophotometer was used to deliver the UV–Visible spectra. Cyanoacetyl-3,5-dimethyl-pyrazole 1, N-(2-aminophenyl)-2-cyanoacetamide 2, and 2-cyano- methyl benzimidazole 3 were prepared according to the literature.<sup>[27]</sup>

### 3.2 | General synthesis procedures

#### 3.2.1 | Synthesis of 1-cyanoacetyl-3,5-dimethylpyrazole (1)

An equimolar amount of 2-cyanoacetohydrazide (10 mmol) was treated with acetyl acetone (10 mmol) in the presence of HCl (catalytic amount) in water (15 ml). The mixture was stirred at room temperature for 2 hr. The white precipitate was filtered off, washed with cold water, and subsequently dried in oven (yield 91%).

### 3.2.2 | Synthesis of N-(2-aminophenyl)-2-cyanoacetamide (2)

Finely powdered o-phenylenediamine (10 mmol) was suspended in toluene (20 ml). 1-Cyanoacetyl-3,5-dimethylpyrazole 1 (10 mmol) was added with stirring. The reaction mixture was stirred at reflux for 15 min. The precipitated product was filtered off and was purified.

### 3.2.3 | Synthesis of 2-cyanomethylbenzimidazole (3)

N-(2-aminophenyl)-2-cyanoacetamide (2) 10 (mmol) was fused above its melting point in an oil-bath for ~15 min. The solid product remained was triturated with methanol. The solid product obtained was filtered and crystallized from ethanol (yield 96%).

### 3.2.4 | N-(4-substituted sulfamoyl phenyl)-1H-benzo[d]imidazole-2-carbohydrazonoylcyanide derivatives (6a–g)

4-Aminobenzenesulfonamide derivatives (4a–g) (10 mmol) were suspended in water (50 ml). HCl (10 ml, 36.5%) was added to this solution by stirring. The mixture was gradually heated up to 70°C till a clear solution was obtained. The solution was cooled to 0–5°C in an ice bath. A sodium nitrite solution (0.5 g in 5 ml H<sub>2</sub>O) was then added over a period of 5 min with stirring. 2-cyanomethylbenzimidazole 3 (10 mmol) was dissolved ethanol in the presence of sodium acetate and kept at 0–5°C. Subsequently, the diazonium salt solution was added with the occasional slow stirring to the 2-cyanomethylbenzimidazole solution for 10–15 min. The formed precipitate was filtered, dried, and recrystallized from proper solvent to give the pure product 6.

### 3.2.5 | N-(4-sulfamoylphenyl)-1H-benzo[d]imidazole-2-carbohydrazonoyl cyanide (6a)

Yellow crystals (84%), m.p. 291°C [Lit<sup>31</sup>.mp 290°C].

### 3.2.6 | N-(4-(N-carbamimidoylsulfamoyl)phenyl)-1H-benzo[d]imidazole-2-carbohydrazonoyl cyanide (6b)

Yellow crystals (89%), m.p. 250°C; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,421, 3,221 (NH), 2,222 (C≡N), 1,681 (C=N), 1,624 (N=N), 1,365, 1,134 (SO<sub>2</sub>); <sup>1</sup>H NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 5.47 (s, 1H,

NH<sub>2</sub>, D<sub>2</sub>O exchangeable), 7.36 (s, 2H, NH, D<sub>2</sub>O exchangeable), 7.35–7.95 (m, 10H, Ar-H + 2NH), 14.68 (s, 1H, NHSO<sub>2</sub>, D<sub>2</sub>O, exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 102.15, 106.26, 110.28, 114.69, 115.26 (CN), 116.73, 124.58, 125.77, 127.75, 128.23, 133.86, 139.48, 144.58 (C=N), 144.95 (C=N imidazole), 145.38 (C=N), 158.06 (NH=C–NH<sub>2</sub>). Calcd; for C<sub>16</sub>H<sub>14</sub>N<sub>8</sub>O<sub>2</sub>S (382.40): %C: 50.25; %H: 3.69; %N: 29.30; %S: 8.38. Found; %C: 50.10; %H: 3.60; %N: 29.40; %S: 8.30.

### 3.2.7 | N-(4-(N-(thiazol-2-yl)sulfamoyl)phenyl)-1H-benzo[d]imidazole-2-carbohydrazonoyl cyanide (6c)

Yellow crystals (85%), m.p. 270°C; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,417, 3,244, 3,186 (NH), 2,218 (C≡N), 1,600 (N=N), 1,323, 1,145 (SO<sub>2</sub>); <sup>1</sup>H NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 5.29, 5.31 (s, 2H, thiazole-H), 7.26–7.86 (m, 9H, Ar-H + NH), 12.76 (s, 1H, NH benzimidazole, D<sub>2</sub>O, exchangeable); 14.68 (s, 1H, NHSO<sub>2</sub>, D<sub>2</sub>O, exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 104.60, 106.46, 108.65, 110.71, 115.37 (CN), 116.29, 124.59, 124.74 (C<sub>3</sub>-thiazole), 124.97, 127.75, 128.26, 136.24, 137.55 (C<sub>2</sub>-thiazole), 137.61, 144.87 (C=N), 145.66 (C=N imidazole), 146.33 (C–NH), 169.15 (C<sub>5</sub>-thiazole). Calcd; for C<sub>18</sub>H<sub>13</sub>N<sub>7</sub>O<sub>2</sub>S<sub>2</sub> (423.47): %C: 51.05; %H: 3.09; %N: 23.15; %S: 15.14. Found; %C: 51.00; %H: 3.10; %N: 23.20; %S: 15.10.

### 3.2.8 | N-(4-(N-(pyrimidin-2-yl)sulfamoyl)phenyl)-1H-benzo[d]imidazole-2-carbohydrazonoyl cyanide (6d)

Yellow crystals (90%), m.p. 302°C [Lit<sup>31</sup>.mp 300°C]; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,417, 3,240, 3,190 (NH), 2,222 (C≡N), 1,581 (N=N), 1,342, 1,157 (SO<sub>2</sub>); <sup>1</sup>H NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 7.01 (s, 1H, pyrimidine-H), 7.35–7.98 (m, 9H, Ar-H + NH), 8.48–8.52 (m, 3H, 2CH-pyrimidine-H + NH benzimidazole), 14.69 (s, 1H, NHSO<sub>2</sub>, D<sub>2</sub>O, exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 102.90, 106.89, 110.22, 114.77, 115.12 (CN), 116.25, 116.58 (C<sub>3</sub>-pyrimidine), 124.60, 125.69, 129.62, 129.98, 134.06, 135.16, 135.79 (C=N), 145.25 (C=N imidazole), 145.94 (C–NH), 157.33 (C<sub>2</sub>, C<sub>4</sub>-pyrimidine), 158.82 (C<sub>6</sub>-pyrimidine). Calcd; for C<sub>19</sub>H<sub>14</sub>N<sub>8</sub>O<sub>2</sub>S (418.44): %C: 54.54; %H: 3.37; %N: 26.78; %S: 7.66. Found; %C: 54.50; %H: 3.30; %N: 26.70; %S: 7.50.

### 3.2.9 | N-(4-(N-acetylsulfamoyl)phenyl)-1H-benzo[d]imidazole-2-carbohydrazonoyl cyanide (6e)

Yellow crystals (85%), m.p. 230°C; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,321, 3,275 (NH), 2,214 (C≡N), 1,716 (C=O), 1,597

(N=N), 1,327, 1,153 (SO<sub>2</sub>); <sup>1</sup>HNMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 1.94 (s, 3H, CH<sub>3</sub>), 5.24 (s, 1H, NH amide, D<sub>2</sub>O exchangeable), 7.34–7.97 (m, 8H, Ar-H), 12.05 (1s, 1H, NH benzimidazole, D<sub>2</sub>O exchangeable), 14.71 (1s, 1H, NHSO<sub>2</sub>, D<sub>2</sub>O exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$  23.67 (CH<sub>3</sub>), 105.86, 107.22, 110.58, 115.27, 115.32 (CN), 116.04, 124.73, 129.74, 130.20, 133.81, 133.96, 136.63, 144.65 (C=N), 146.32 (C=N imidazole), 146.93 (C–NH), 169.42 (C=O). Calcd; for C<sub>17</sub>H<sub>14</sub>N<sub>6</sub>O<sub>3</sub>S (382.40): %C: 53.40; %H: 3.69; %N: 21.98; %S: 8.38. Found; %C: 53.50; %H: 3.69; %N: 21.90; %S: 8.20.

### 3.2.10 | *N*-(4-(*N*-(5-methylisoxazol-3-yl)sulfamoyl)phenyl)-1*H*-benzo[*d*]-imidazole-2-carbohydrazonoyl cyanide (6f)

Yellow crystals (86%), m.p. 220°C; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,213, 3,163, (NH), 2,233 (C=N), 1,612 (N=N), 1,334, 1,165 (SO<sub>2</sub>); <sup>1</sup>HNMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 2.3 (s, 3H, CH<sub>3</sub>), 6.16 (s, 1H, isoxazole-H), 7.37–7.96 (m, 9H, Ar-H + NH), 11.41 (s, 1H, NH benzimidazole, D<sub>2</sub>O exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 12.49 (CH<sub>3</sub>), 104.52, 107.14, 110.37, 115.03, 115.63 (CN), 116.62, 124.71, 125.25, 128.91, 129.39, 134.32, 135.37, 144.65 (C=N), 145.82 (C=N imidazole), 145.89 (C–NH), 146.48 (C<sub>4</sub>-isoxazole), 157.97 (C<sub>3</sub>-isoxazole), 170.85 (C<sub>5</sub>-isoxazole). Calcd; for C<sub>19</sub>H<sub>15</sub>N<sub>7</sub>O<sub>3</sub>S (421.44): %C: 54.15; %H: 3.59; %N: 23.27; %S: 7.61. Found; %C: 54.10; %H: 3.60; %N: 23.20; %S: 7.60.

### 3.2.11 | *N*-(4-(*N*-(4-methylpyrimidin-2-yl)sulfamoyl)phenyl)-1*H*-benzo[*d*]-imidazole-2-carbohydrazonoyl cyanide (6g)

Yellow crystals (80%), m.p. 281°C [Lit<sup>31</sup>.mp 280°C]; IR (KBr) cm<sup>-1</sup>:  $\nu_{\max}$  3,390, 3,228 (NH), 2,225 (C=N), 1,597 (N=N), 1,369, 1,157 (SO<sub>2</sub>); <sup>1</sup>HNMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$ : 2.33 (s, 3H, CH<sub>3</sub>), 6.04 (s, 1H, NH, D<sub>2</sub>O exchangeable), 6.92 (s, 1H, pyrimidine-H), 7.31–7.94 (m, 9H, Ar-H + NH benzimidazole), 8.32, 8.34 (2s, 2H, pyrimidine-H), 14.67 (1H, s, NHSO<sub>2</sub>, D<sub>2</sub>O, exchangeable); <sup>13</sup>C NMR (DMSO-*d*<sub>6</sub>, ppm)  $\delta$  23.69 (CH<sub>3</sub>), 102.88, 106.72, 110.14, (2)114.76, 115.17 (CN), 116.27 (C<sub>3</sub>-pyrimidine), 124.66, 125.79, (2)129.87, (2)134.06, 136.03 (C=N), 145.35 (C=N imidazole), 145.72 (C–NH), 156.82 (C<sub>4</sub>-pyrimidine), 157.78 (C<sub>2</sub>-pyrimidine), 168.95 (C<sub>6</sub>-pyrimidine). Calcd; for C<sub>20</sub>H<sub>16</sub>N<sub>8</sub>O<sub>2</sub>S (432.46): %C: 55.55; %H: 3.73; %N: 25.91; %S: 7.41. Found; %C: 55.40; %H: 3.70; %N: 25.80; %S: 7.40.

## 3.3 | Antibacterial activity

The antibacterial activities of novel sulfonamide derivatives (**6a–g**) were tested against *S. aureus* (RCMB 010010), *S. epidermidis* (RCMB 009–2), *E. coli* (RCMB 010052), and *K. pneumonia* (003-1) using Hole-plate diffusion method. All bacterial stains were provided from Regional Center for Mycology and Biotechnology (RCMB), Al-Azhar University, Cairo, Egypt.<sup>[41]</sup> 6 mm holes were made in Mueller-Hinton agar plate, which was previously seeded with tested bacteria (10<sup>8</sup> CFU/ml). 100  $\mu$ l of the tested compound (5 mg dissolved in ml dimethylsulfoxide [DMSO]) was filled in the hole, and after that, the plates were incubated for 24 hr at 37°C. After incubation, the inhibition zones were measured in mm. DMSO was embedded in each experiment as a negative control. The experiment was performed in triplicate. For determining the MIC of active compounds, each active compound was further investigated using pervious method against the selected bacteria (twofold serial dilutions of each active compound were tested). The lowest concentration, which inhibiting the growth of bacteria after the incubation period was determined as the MIC.

## 3.4 | Gel formulation design

The novel sulfonamide derivative (**6d**) was dissolved in an appropriate amount of DMSO, then propylene glycol was added, and the solution was stirred for 10 min. The solution was stirred with the aqueous dispersion of CRB. TEA was added to neutralize the CRB solution. The final weight of the gel was adjusted to 100 g with distilled water. The different samples were prepared according the following recipe (Table 3).

## 3.5 | Gel characterization

### 3.5.1 | Physical appearance and homogeneity

The physical appearance and homogeneity of the prepared gels were tested by visual observations.

### 3.5.2 | pH

Since the gel's pH must be ideally near to normal pH of the skin to avoid any irritation, pH measurement of the gel was carried out by dipping the glass electrode



**TABLE 3** Recipe of gel formulations

| Gel formulation code | Carbopol (g) | Triethanolamine (ml) | Solvent (ml) |
|----------------------|--------------|----------------------|--------------|
| Blank 1              | 0.5          | 1.5                  | —            |
| Blank 2              | 1            | 1                    | —            |
| Blank 3              | 1            | 1.5                  | —            |
| Blank 4              | 1            | 1.5                  | 2            |
| Blank 5              | 1            | 2                    | —            |
| Blank 6              | 1.5          | 1.5                  | —            |
| Blank 7              | 2            | 1.5                  | —            |
| <b>6d</b> gel        | 1            | 1.5                  | 2            |
| ES gel               | 1            | 1.5                  | 2            |
| <b>6d</b> -ES gel    | 1            | 1.5                  | 2            |

completely into the gel system to cover the electrode. The results are the mean of three readings.

### 3.5.3 | Viscosity

Viscosity of the gel was determined using Brookfield viscometer (DV2T Viscometer, spindle no. 27). All measurements were done in triplicate at room temperature at a speed of 0.1 rpm.

### 3.5.4 | Spreadability

Spreadability was determined using an apparatus suggested by Mutimer,<sup>[42]</sup> which was suitably modified in the laboratory and used for the study. It consists of a wooden block provided by a pulley at one end. A ground glass slide was fixed on this block. About 1 g of gel under study was placed on this slide. Another slide with one end tied to weight of 50 g was placed on the top of the gel, such that the gel was sandwiched between the two slides in an area occupied by a distance of 7.5 cm along the slides. A weight of 500 g was placed on the top of the two slides for 5 min to expel air and to compress the sample to a uniform thickness. The weight was removed, and the excess of the gel was scrapped off from the edges. The upper slide was then subjected to a pull of the 50 g weight and the time taken by it to separate completely from the fixed slide was recorded. The gel spread ability was evaluated through the following equation:

$S = ML/T$  where, S = spreadability. M = weight (g) tied to upper slide. L = length (cm) of glass slide. T = time (s) taken by the slide to separate.

### 3.5.5 | Skin irritation

A total of 100 mg gel was applied on an area of 2 cm for 6 hr, on the interior surface of upper arm and covered with cotton bandage. After 6 hr, the examination for the presence of lesion or irritation was done and readings are made according to the scale given by Draize<sup>[43]</sup>; No irritation: 0; Slight irritation: 1; Irritation: 2.

### 3.6 | Collection of ES

Excretion/Secretion was collected from *L. cuprina* maggots that obtained from maintained culture at Medical Entomology laboratory at Animal House, Faculty of Science, Al-Azhar University. Briefly, the maggots were washed, sterilized, and then incubated with phosphate buffered saline for 6 hr at 25°C. ES was centrifuged, filtrated, and stored at −20°C.<sup>[44]</sup>

### 3.7 | Wound healing activity

The wound healing activity of the novel sulfonamide derivatives (**6d**) was assessed in male Wister rats (200–250 g), the rats were obtained from the Animal House, Faculty of Science, Al-Azhar University. The rats were adapted in cages for 2 weeks with provided food and water. By Ketamine 50 mg/kg, the rats were anesthetized, and then shaved the skin. A circle excision (25 mm in diameter) was made on the dorsal skin of rats. Randomly, the rats were divided into five groups; Control, negative control, three treated groups (**6d**, ES, and **6d**-ES), and three rats in each group. The negative control, **6d**, ES, and **6d**-ES groups were applied with blank gel

and the sulfonamide derivative (**6d**) gel, ES gel, and **6d**-ES combined gel formulations, respectively. The wound was imaged, and its area was measured by ImageJ software. The wound contraction was calculated using Equation (1).<sup>[29]</sup>

$$\text{Wound contraction (\%)} = \frac{A_0 - A_t}{A_0} \times 100 \quad (1)$$

where  $A_0$  and  $A_t$  are the wound area at initial time and time  $t$ , respectively.

## 4 | CONCLUSIONS

A series of *N*-(4-substituted sulfamoylphenyl)-1*H*-benzo[d]imidazole-2-carbohydrazonoyl-cyanide derivatives (**6a–g**) were synthesized in good yields by coupling of diazotized 4-aminobenzenesulfonamides (**4a–e**) with 2-cyanomethylbenzimidazole (**3**) in the presence of sodium acetate. These derivatives (**6a–g**) were tested against some strains of bacteria, and the **6d** derivative showed the best antibacterial activity. In vivo, the wound healing activity of **6d** was investigated in a gel form. The sulfonamide derivative delayed the wound contraction, while its activity was enhanced by combination with ES of *L. cuprina* maggots. The wound healing activity of sulfonamides needs further investigation with different concentrations of ES and other similar compounds to enhance the activity of sulfonamides in topical applications.

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